Mr Sheharyar

Computer Science and Information Systems Honours Graduate

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Professional Summary

I am an Honours graduate in Computer Science and Information Systems with a strong foundation in programming, object-oriented design, databases, and systems development. Currently working as a Software Developer at Wirk, where I design and implement backend services, work with databases, and build APIs for real-world applications while growing my technical knowledge.

I'm passionate about AI engineering and machine learning, with a focus on building intelligent systems for real-world use. I regularly work on personal projects and continuously develop my skills to stay up to date with the latest advancements.

Experience

Software Engineer & Game developer, *Wirk (Pty) Ltd* 01/2026 – Present
Port Elizabeth

- Worked as part of the AI Research & Development team, researching, prototyping, and implementing AI features.
- Used Python for backend development, scripting, and integration with existing systems.
- Developed Gameplay features for an existing live game using Unreal Engine 5, working with both C++ for subsystems and Blueprint for visual scripting.
- Implemented three custom game launchers, handling game distribution, key generation, subscription management, and authentication (OAuth)
- Built an GraphQL backend and serverless functions in .NET to support application services.
- Contributed to frontend applications and WordPress-based systems.
- Worked with PostgreSQL databases and implemented CI/CD pipelines using Azure DevOps to automate deployment processes.

Tech Lead, *Nelson Mandela University* 03/2025 – 11/2025

- Provided technical guidance and support to student teams developing web, mobile, and desktop applications, assisting with architecture design, debugging, and technology stack selection.

Data Structures and Algorithms Tutor, *Freelance* 01/2023 – Present

- Tutored over 20 students in core data structures, improving their grades by an average of 10%.
- Assisted students in debugging and optimizing their code, teaching best practices for writing efficient, clean, and maintainable code.
- Assisted students with coding assignments, tests, and exam preparations, offering targeted coding practice and problem-solving strategies.

IT Technician, *CellZone* 01/2021 – 02/2022

- Engaged with customers, promptly addressing inquiries related to mobile phones, computer software, and hardware.
- Responsible for repairing and upgrading mobile devices, as well as installing and upgrading computer hardware and software

Skills

Coding Languages
Java, C#, JavaScript, Python, SQL

Machine learning Skills - From coursework
Feedforward, Convolutional, Recurrent Neural Networks, Autoencoders, Transformers; Evolutionary Algorithms; Model training & optimization

Frameworks
React.js, React Native, Node.js, ASP.NET, Django, Blazor, SpringBoot, Vite, JUnit 5, Tailwind CSS, Unity, ARCore, TorchSharp

Developer Tools
Git, Visual Studio, IntelliJ, Android Studio, WebStorm, Microsoft Visio.

Database
MySQL, PostgreSQL, MongoDB, Firebase, Supabase

Soft Skills
Public Speaking, Problem Solving, Team Leadership, Communication, Time Management

Certifications

The Complete 2024 Web Development Bootcamp - Udemy
A 60+ hour full-stack development course, covering modern technologies like JavaScript (ES6), React, Node.js, Express, MongoDB, and more, with a focus on building real-world projects and deploying scalable web applications.

Education

Honours Degree - Computer Science and Information Systems, *Nelson Mandela University* 01/2025 – 12/2025
Grade: Cum Laude South Africa
Relevant Coursework:
Evolutionary Computing & Intelligent Systems, Virtual Reality Environment Development, Data warehousing, Functional Programming, Usability Engineering, Project Management, Electronic Commerce, Environmental Information Systems.
Thesis: (PRIMED - Project Reporting, Insights, and Module Efficiency Dashboard).

Bachelor's Degree - BCom in Computer Science and Information Systems (Double Major with a Minor in Business), *Nelson Mandela Metropolitan University* 02/2022 – 12/2024
South Africa
Grade 70
Relevant coursework: Advanced Programming, Advanced Data Structures, Data Structures and Algorithms, Database Systems, Computer Architecture, Web Systems, User Interface Design, CS&IS Project, Business Statistics, Mathematics, Business Systems, Management Information Systems, General & Strategic Management, Financial Management.

Projects

Movie & TV Show Discovery App, React Native / Expo 12/2025 – 01/2026
- Developed a cross-platform mobile app for movie and TV discovery using React Native and Expo
- Integrated TMDB API to fetch and display searchable entertainment content with infinite scrolling
- Integrated Google OAuth 2.0 sign-in with Appwrite for secure user authentication
- Used cloud database to store and manage user favorite movies and TV shows
- **Tech stack: React Native(Expo), Tailwind**

PRIMED - Project Reporting, Insights, and Module Efficiency Dashboard, Honours Project 02/2025 – 11/2025
- Developed a web-based dashboard for Nelson Mandela University to streamline module coordination tasks.
- Implemented automated team creation and supervisor/peer review allocation.
- Built milestone tracking and mark calculation features with real-time updates.
- Provided team and student insights through analytics dashboards.
- Integrated automated email notifications for deadlines and updates.
- **Tech Stack: React.js, Supabase, Tailwind CSS**

UNI_FEVER, 3rd year Computer Science Final Project 02/2024 – 11/2024
- Project Overview: Collaborated to develop a full stack Student Notes document repository for my final year project.
- User Functionality: Allows university students to upload, view, download, and bookmark pdf documents.
- Version Control: Utilized GitHub for version control and collaborative development.
- API integration: Enabled users to interact with PDF documents via a chat bot interface
- Awards: **Best web system** voted by industry partner and **best user interface** voted by students
- **Tech stack: ASP.NET**

CCTV Human vs Non-Human Classification, TorchSharp 09/2025 – 10/2025
- Collaborated to develop an end-to-end deep learning classification pipeline in TorchSharp on 9,858 128x128 RGB images.
- Tested and compared multiple CNN architectures: custom 2-layer CNN, LeNet-5 (classic & modernized), AlexNet, ZFNet, and ResNet variants.
- Achieved highest validation accuracy of 92.52% with custom CNN; other top results: ZFNet 91.38%, AlexNet 89.96%, modernized LeNet-5 89.76%, ResNet 89.57%, classic LeNet-5 66.13%.
- Optimized training using BCE loss, dropout, batch normalization, and learning rate tuning.

Augmented Reality- Fishing, Unity 09/2025 – 10/2025
- Built a mobile AR fishing prototype using ARCore, with plane tracking to anchor a 3D pond and spawn rod/fish entities on detected surfaces.
- Implemented gesture and motion interactions: tap to place pond, swipe to cast, on-screen reel control, and device tilt for rod tension/line control via accelerometer and gyroscope.
- Developed real-time bite/hook logic with proximity triggers, hit detection, scoring, and timed challenge mechanics.
- Created fish Al using finite state machines (FSMs) for patrol, investigate, bite, and flee behaviors, with steering reacting to lure proximity, line tension, and timers.

Virtual reality - Hostage Rescue: Zombie Outbreak, Unity 08/2025 – 09/2025
- Built a mobile VR experience with gated doors, resource pickups, and HUD updates for health, ammo, and keys.
- Implemented NavMesh enemy Al for roaming, detection, chasing, and ranged combat, plus damage systems for zombies and landmines.
- Added comfort locomotion: walk, sprint, jump, snap rotation, and instant 180° turns, with full gamepad controller support.
- Produced a tense survival experience with NPC clue dialogs, environmental VFX, and SFX for footsteps, gunfire, and explosions